

- : SUPER Conductivity:-

Super conductor:- Super conductor is the state of metal in which electric resistance is practically vanish and current flowing without voltage.

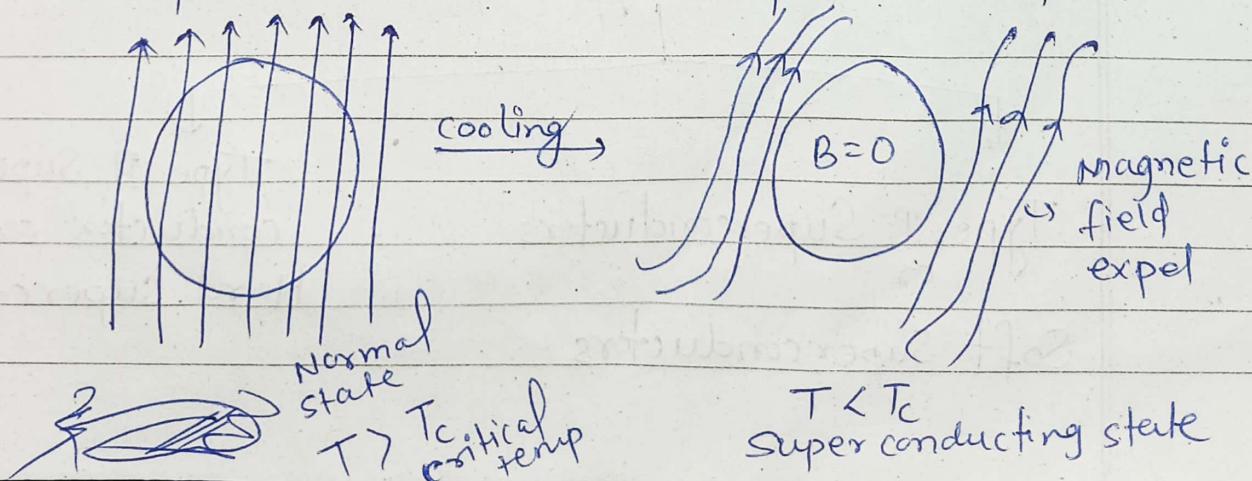
Super conductivity:- The phenomenon in which the electrical resistance of a given specimen of metal or alloy falls suddenly to zero or cooling it to extremely low temp is called Super conductivity.

Transition temperature:- Temperature at which metal converted to Super conductor.

Meissner's effect of flux exclusion

This phenomenon was discovered by german physicist walther Meissner and Robert ochsefeld in 1933

Statement:- When Superconductors are cooled down below the critical temp they expel magnetic field and do not allow the magnetic field to penetrate inside them. this phenomenon in Superconductor is called Meissner effect.



Magnetic induction inside the superconductivity

$$B = 0$$

M = Intensity of magnetisation by magnetising field (H)

The flux density (B) in magnetic material is related to the strength of

$$B = \mu_0 (H + M) = 0 \quad \text{--- (1)}$$

Magnetic Susceptibility

$$\chi_m = \frac{M}{H} = -1$$

from eqn (1)

$$H = -M$$

for diamagnetic material $\chi_m \ll 1$

for Superconductors $\chi_m = -1$

Superconductors

Classification of Superconductors

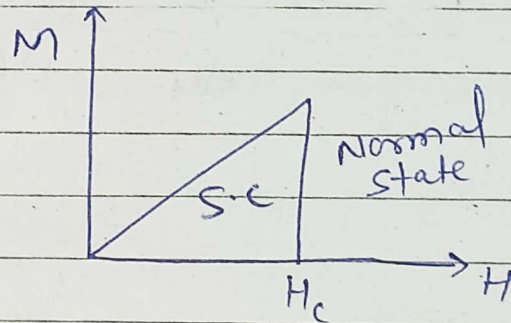
Type-I Superconductors
or

Soft Superconductors

Type-II Superconductor or
Hard Superconductors

Type-I Superconductor:-

- These Superconductors which can tolerate impurities without affecting the Superconducting properties.
- Exhibits perfect and complete Meissner effect
- Behave as a perfect diamagnetic material
- There is only one H_c
- Sudden loss of magnetization
Ex:- Pb, Sn, Hg



Type-II Superconductors

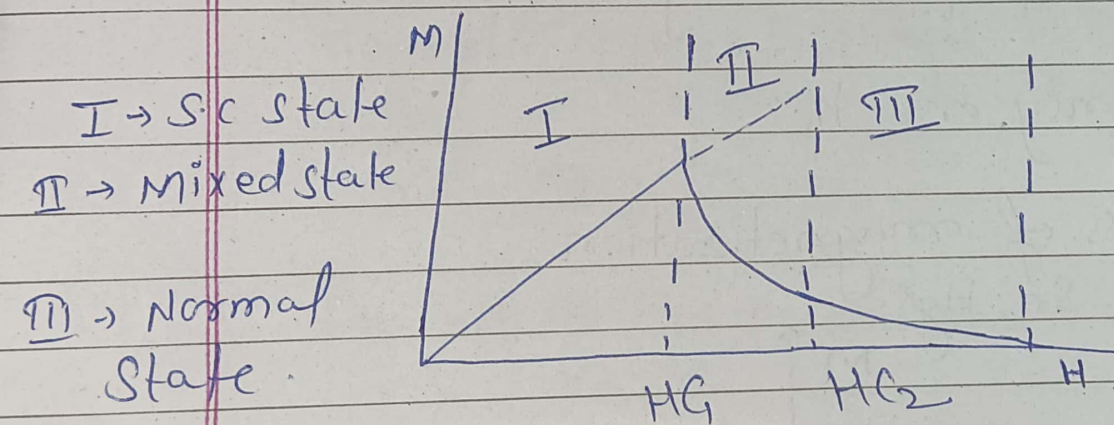
- Those Superconductors which cannot tolerate impurities i.e., impurity affects the Superconducting properties.
- Don't exhibit complete Meissner effect
- Does not behave as a perfect diamagnetic material

- There are two critical magnetic field

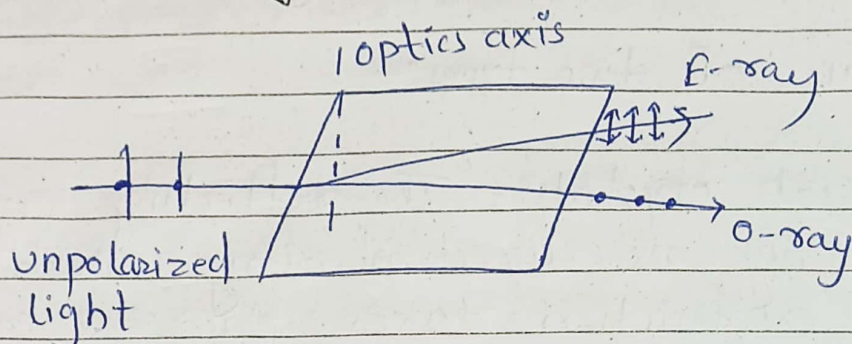
$H_{C1} \rightarrow$ lower CMF

$H_{C2} \rightarrow$ upper CMF

- Gradual loss of magnetisation
Ex:- Nb-Sn, Nb-Ti



W Polarization by double refraction



Polarization by double refraction is the phenomenon by which unpolarized light splits in two components and ray obey the law of refraction is called ordinary ray other ray which does not obey the law of refraction is called Extra-ordinary ray.

The crystal such as calcite in which double refraction takes place is called double refracting crystals. Both the rays are plane polarized. But plane of polarization are $\perp$ to each other.

→ Geometry of calcite crystal

① It is double refracting crystal.

It is colourless & transparent made up of CaCO_3 .

Through it visible as well as

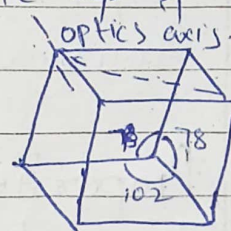
ultraviolet light can pass. There

is a direction in calcite crystal

in which no double refraction takes

place is called optic axis. The plane in which optic axis lies and $\perp$ to the opposite faces of crystal is called

principal plane.



→ Types of double refracting crystals

They are of two types:-

- ① Uni-axial crystals:- The reflecting crystals having only one direction along which no double refraction takes place are called uni-axial crystals.

EX:- calcite, quartz, & etc

- (2) Bi-axial crystals:- The double refracting crystals having two directions along which no double refraction takes place are called Bi-axial crystals.

EX:- mica, Borax, Topaz etc.

→ Double refracting crystal are two types:-

1. Negative crystals
The double refracting crystals like calcite, tourmaline etc.

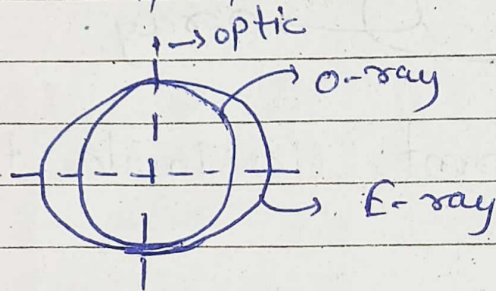
2. Positive crystals
The double refracting crystal like quartz, ice etc.

Properties

1. The velocity of o-ray is same in all dirⁿ.

1. The double refracting velocity of o-ray is same in all directions.

2. The velocity of E-ray is different in different directions. The velocity of E-ray is same as velocity of o-ray along optic axis and is minimum. The velocity of E-ray is maximum along $\perp$ dirⁿ of optic axis.

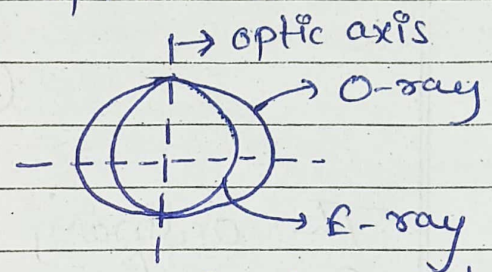


$$\mu_o = \frac{c}{v_o}$$

$$\mu_E = \frac{c}{v_E}$$

So $\boxed{\mu_o > \mu_E}$

2. The velocity of o-ray is different in different directions. It is equal to the velocity of o-ray along optical axis but is maximum. The velocity of E-ray is minimum along $\perp$ direction of optic axis.



3. The refractive index of E-ray is greater than refractive index of o-ray

$$\boxed{\mu_E > \mu_o}$$

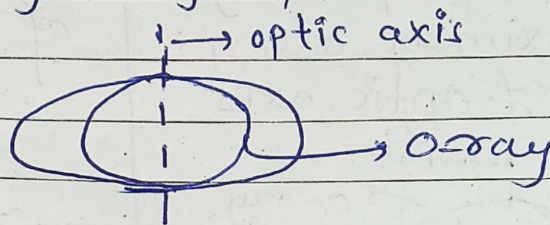
→ Hugen's Theory of double refraction:-

- (1) Each point in double refracting crystal becomes the source of disturbance and sends out the secondary O and E wavefronts.
- (2) The velocity of o-ray is same in all directions, so, the wavefront of o-ray is spherical.

3. The velocity of E-ray is different in directions and so the wavefront of E-ray is ellipsoid.

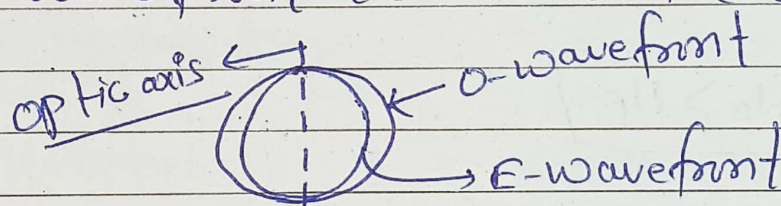


4. In negative crystals the velocity of E-ray is different in different directions. It is equal to o-ray along optic axis and minimum



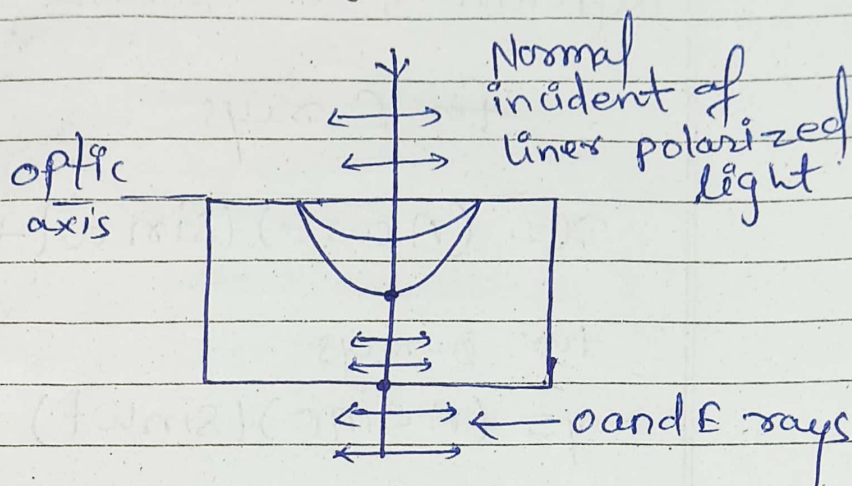
The ordinary wavefront lies inside the E-wavefront

5. In positive crystals. The velocity of E-ray is different in different directions and it is equal to the velocity of o-ray is maximum, so, E-wavefront lies inside the o-wavefront



O and E wavefront intersect along optic axis.

Theory of elliptically, circularly polarized light



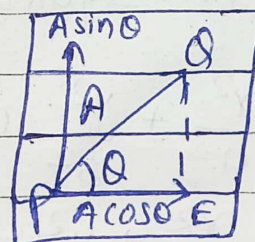
When linearly polarized light incident normally in a double refractive crystal whose optic axis is $\parallel$ to the refracting face, then it splits up in O and E-rays.

O-wavefront is spheroid and E-wavefront is ellipsoid. The O and E wavefront cut along optic axis. On emergence these rays are elliptically polarized, circularly polarized or plane polarized light, depending on the thickness of crystal and on direction of incident amplitude with optic axis.

Theory:- Let A is the amplitude of incident ray which makes angle θ with optic axis. It acts along PQ.

Resolve A in two $\perp$ components

- (i) $A \cos \theta$ for E-rays
- (ii) $A \sin \theta$ for O-rays



on emergence then there will be phase difference ϕ between O and E-rays.

for E-rays

$$x = (A \cos \theta) (\sin \omega t + \phi)$$

for O-rays

$$y = (A \sin \theta) (\sin \omega t)$$

$$\text{put } A \cos \theta = a \quad A \sin \theta = b$$

$$x = a \sin(\omega t + \phi)$$

$$y = b \sin \omega t$$

$$\frac{x}{a} = \sin \omega t \cos \phi + \cos \omega t \sin \phi$$

$$\frac{y}{b} = \sin \omega t$$

$$\frac{x}{a} = \frac{y}{b} \cos \phi + \sqrt{1 - \frac{y^2}{b^2}} \sin \phi$$

$$\cos^2 \omega t = 1 - \sin^2 \omega t$$

$$\cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

$$\frac{x}{a} - \frac{y}{b} \cos \phi = \sqrt{1 - \frac{y^2}{b^2}} \sin \phi$$

on squaring both sides

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \phi - \frac{2xy \cos \phi}{ab} = \left(1 - \frac{y^2}{b^2}\right) \sin^2 \phi$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy \cos \phi}{ab} = \sin^2 \phi \quad \text{--- (1)}$$

This is eqn of ellipse so,

emergent light is elliptically polarized

Special cases:

- ① If the thickness of crystal is such that $\phi = 0$

$$\cos \phi = \cos 0 = 1$$

$$\sin \phi = 0 \text{ eqn (1) gives}$$

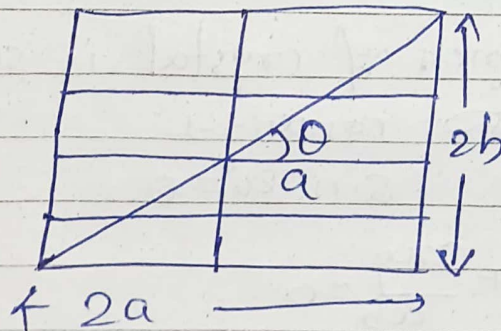
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

$$\left(\frac{x}{a} - \frac{y}{b}\right)^2 = 0$$

$$\frac{x}{a} - \frac{y}{b} = 0$$

$$\frac{y}{b} = \frac{x}{a} \Rightarrow \boxed{y = \frac{b}{a}x}$$

It is the eqn of straight line so, emergent light ray is linearly polarized with slope b/a .



②

If the thickness of crystal is such that $\phi = 90^\circ$

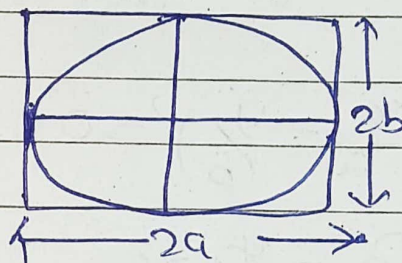
$\sin 90^\circ = 1$, $\cos 90^\circ = 0$ eq (1) gives

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \quad \text{--- (3)}$$

It is the eqn of Symmetrical ellipse, whose major and minor axis coincide with the co-ordinate axis.

If $a = b$
eqn (3) gives

$$\boxed{x^2 + y^2 = a^2}$$



It is eqn of circle. So the emergent is circularly polarized light.

$\therefore a = b$

$A \cos 0 = A \sin 0$

$\tan 0 = 1 \quad \boxed{0 = 45^\circ}$

3. If the thickness of crystal is such as $\phi = 180^\circ$

~~cos 0~~ $\cos 180^\circ = -1$

eq (1) gives $\sin 180^\circ = 0$

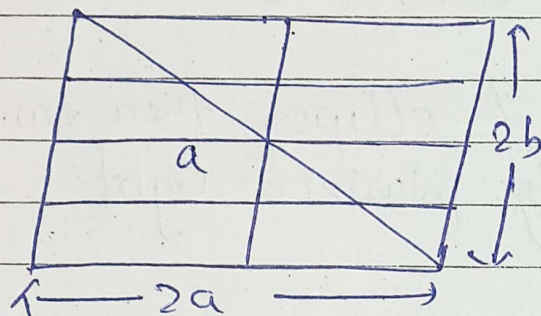
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{ab} = 0$$

$$\left(\frac{x}{a} + \frac{y}{b} \right)^2 = 0$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0 \Rightarrow y/b = -x/a$$

$$y = -\frac{b}{a}x \quad \text{--- (4)}$$

So the emergent light is linearly polarized with Slope $-\frac{b}{a}$



④ When $\phi = 45^\circ$ then

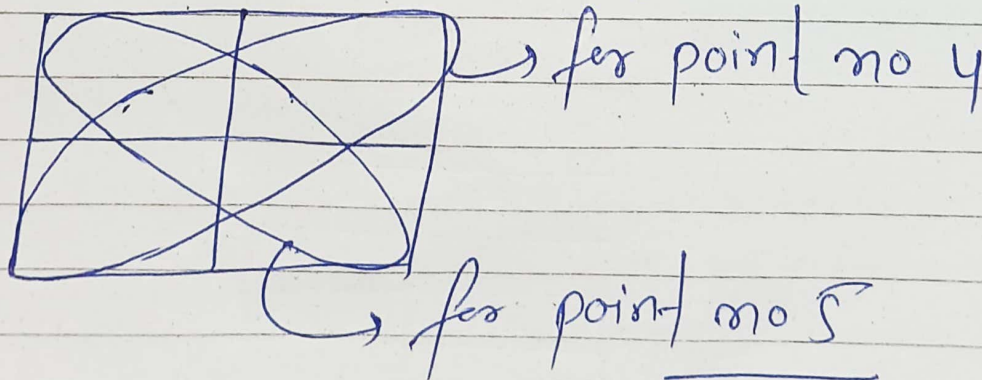
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{\sqrt{2}ab} = \frac{1}{2}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

⑤

It is the eqn of oblique ellipse so the emergent light is elliptically polarised light.



⑤ If $\phi = 135^\circ$, $\sin \phi = \frac{1}{\sqrt{2}}$
 $\cos \phi = -\frac{1}{\sqrt{2}}$

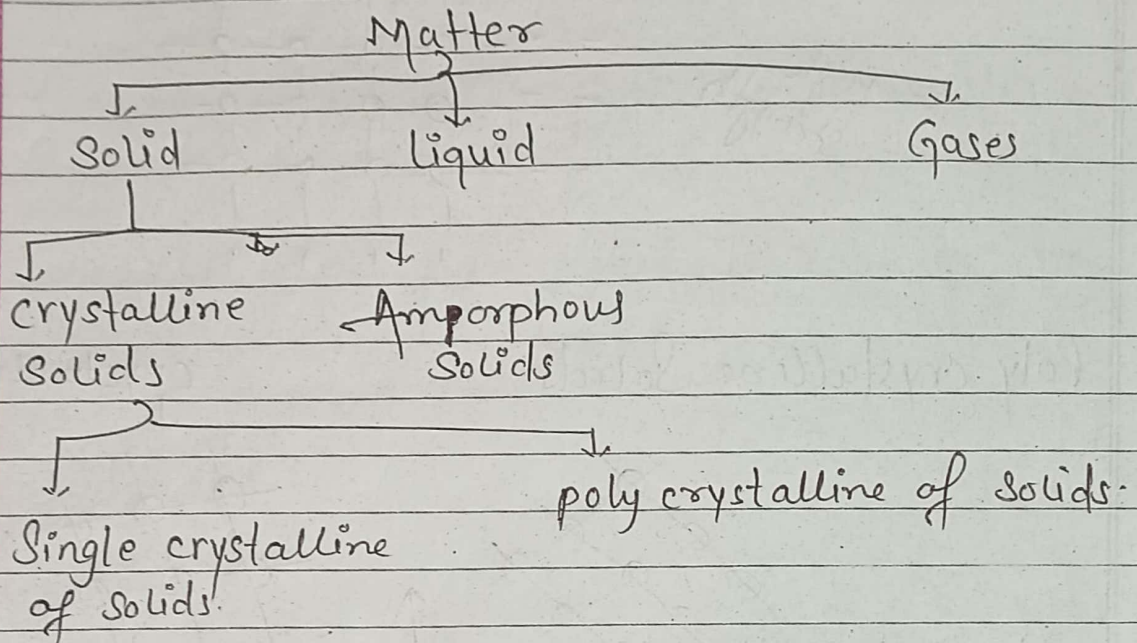
eqn ① gives

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{\sqrt{2}ab} = \frac{1}{2} \quad \text{--- ⑥}$$

It is eqn of ellipse then emergent light is elliptically polarized light.

∴ Solid state:-

Solid state:- A branch of physics deals with physical properties of solids.



Crystalline Solids:-

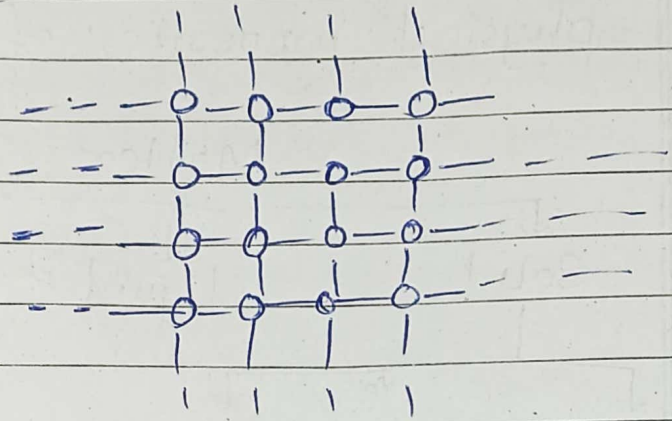
- Atoms or molecules regularly and periodically.
- ~~large~~ ^{long} range order crystal
- Sharp melting point.
- low energy state
- Anisotropic in nature.

Amorphous :- Atoms or molecules are regularly arranged.

- Short range order
- low melting point
- High energy state
- Isotropic in nature

Single crystalline solids:-

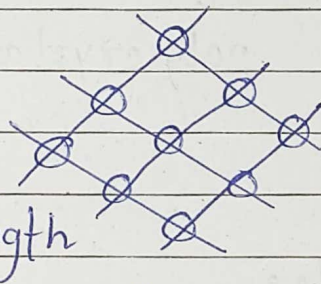
High strength



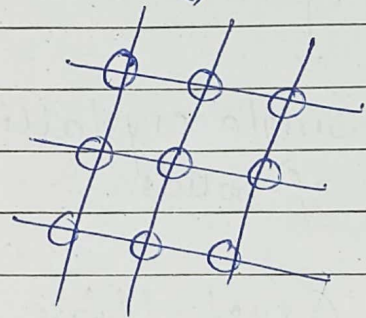
Poly crystalline Solids

crystallite

low strength



crystallite Grain boundary



→ Some important definition

(1) Lattice or crystal lattice:-

Represents arrangement of atoms or molecules in plane or space

• • • • ← lattice point

• • • • ← lattice

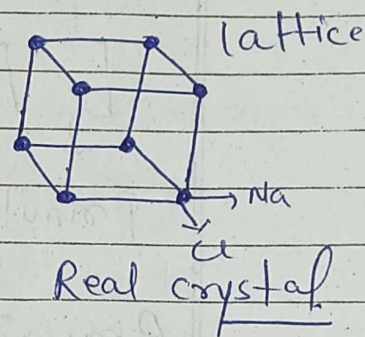
Every lattice point identical surroundings.

Plane lattice:- Arrange lattice point in plane (2D)

Space lattice:- Arrange lattice point in Space (3D)

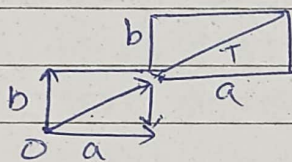
Basis:- Group of atoms or molecules attach with each lattice point to give a real crystal.

Crystal structure = lattice + Basis



Unit cell:-

Smallest part of plane Space lattice which can construct entire crystal by translation repetition.



Plane lattice

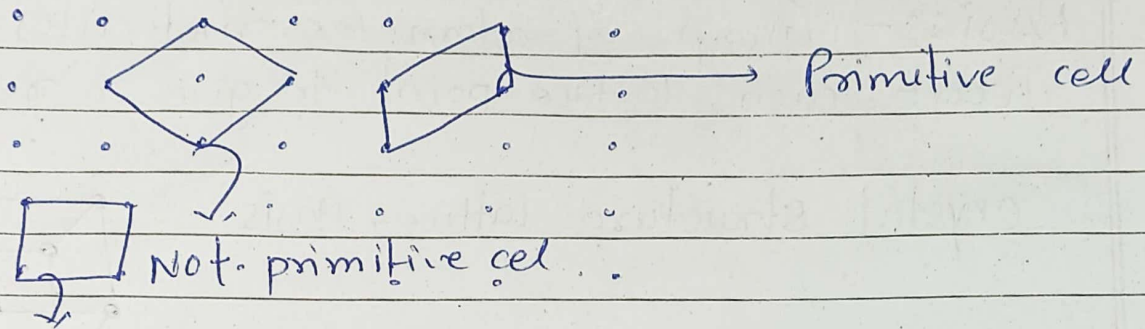
Unit cell = Smallest area

$\vec{a}, \vec{b}$ = basis vector or Translation vector
 $\vec{T}$ = Translation vector

Area of unit cell = $|\vec{a} \times \vec{b}|$

Space lattice, $\vec{a}, \vec{b}, \vec{c}$ = basis vector
volume of unit cell $|\vec{a} \times \vec{b} \cdot \vec{c}|$

Primitive cell:- Primitive cell is a type of unit cell which contains only one lattice point.

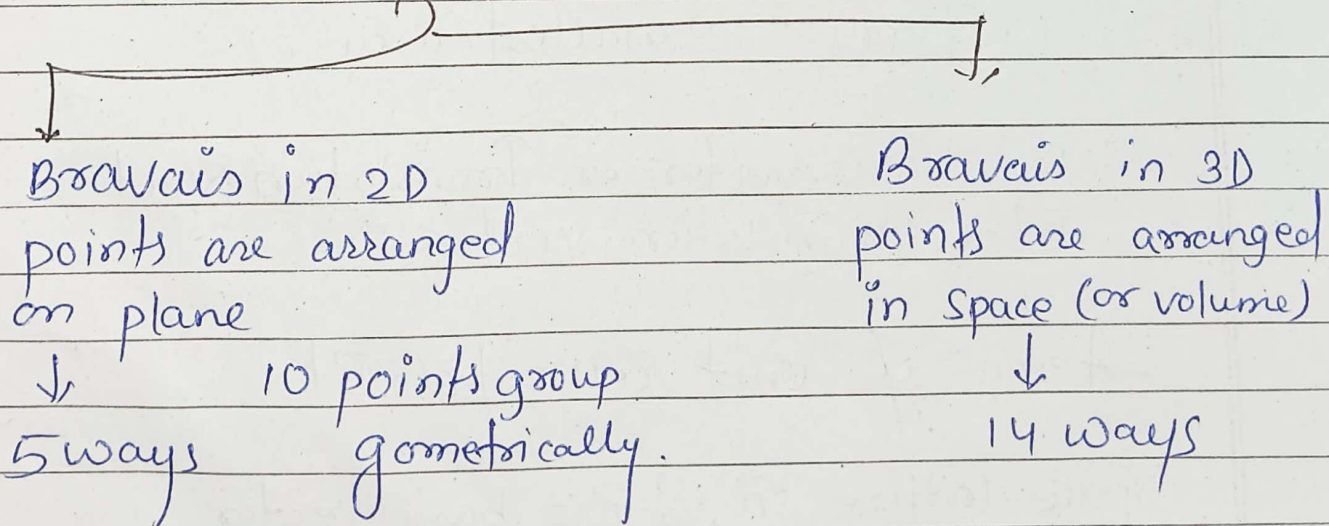


Primitive cell

Bravais lattice:-

There are various ways of positioning points in space such that all points have identical surroundings. There are known as Bravais lattice.

Bravais lattice



Braavais lattices are invariant under symmetry operations.

No of atoms in unit cell

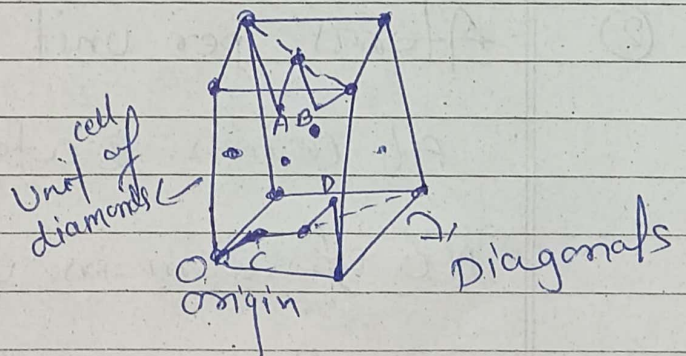
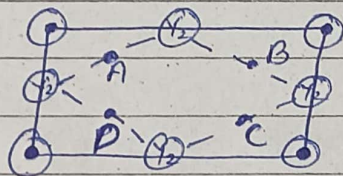
$$N = \frac{N_c}{8} + \frac{1}{2} + N_b$$

Co-ordination no:- No of nearest neighbour atoms of a atom present in unit cell.

Gives the information of packing of atoms.

Diamond crystal structure:-

- Diamond crystal structure is supposed to be a face crystal lattice with 4 additional carbon atoms.



- ① Co-ordination no
- ② Atoms per unit cell
- ③ lattice constant (a)
- ④ Packing fraction

Position of 4 additional carbon atoms

$$C \left(\frac{a}{4}, \frac{a}{4}, \frac{a}{4} \right) \text{ or } C \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)$$

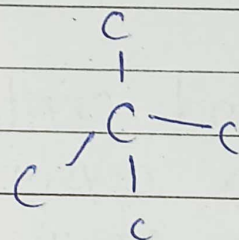
$$D \left(\frac{3a}{4}, \frac{3a}{4}, \frac{a}{4} \right) \text{ or } D \left(\frac{3}{4}, \frac{3}{4}, \frac{1}{4} \right)$$

$$A \left(\frac{a}{4}, \frac{3a}{4}, \frac{3a}{4} \right) \text{ or } A \left(\frac{1}{4}, \frac{3}{4}, \frac{3}{4} \right)$$

$$B \left(\frac{3a}{4}, \frac{a}{4}, \frac{3a}{4} \right) \text{ or } B \left(\frac{3}{4}, \frac{1}{4}, \frac{3}{4} \right)$$

① Co-ordination no:-

$$\boxed{C.N = 4}$$



② Atoms per unit cell

At corner 8 atom

$$\text{No of atom in unit cell} = 8 \times \frac{1}{8} = 1$$

$$\text{No of atom at faces} = 6 \times \frac{1}{2} = 3$$

Additional

→ Total no of atoms in Diamond unit cell = 8

(3) lattice constant (a)

$$r = \frac{\sqrt{3}a}{8}$$

$$a = \frac{8r}{\sqrt{3}}$$

(4) Packing fraction

$$P.F = \frac{\text{Volume occupied by atoms}}{\text{Volume of unit cell}}$$

$$\begin{aligned}\text{Volume occupied by a atom} &= 8 \times \frac{4}{3} \pi r^3 \\ &= \frac{32}{3} \pi r^3\end{aligned}$$

$$\begin{aligned}\text{Volume of unit cell} &= a^3 = \left(\frac{8r}{\sqrt{3}}\right)^3 \\ &= \frac{64 \times 8 \times r^3}{3\sqrt{3}}\end{aligned}$$

$$\begin{aligned}P.F &= \frac{8 \times \frac{4}{3} \times \pi r^3}{\frac{64 \times 8 \times r^3}{3\sqrt{3}}} = \frac{4\pi \times \sqrt{3}}{64 \times 16} = \frac{\sqrt{3} \pi}{16} \\ &= 0.34\end{aligned}$$

$$\boxed{\% PF = 34\%}$$

Diamond atoms